WORKED SOLUTION

# STEP III 2019, Question 8

Pure • Vectors and 3D Geometry

**STEP III 2019, Question 8**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

## The key idea

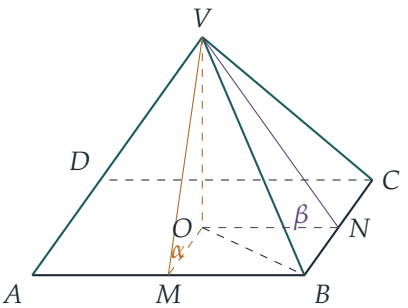
Put coordinates on the pyramid. Each sloping face contains a base edge, which is horizontal, so its normal is perpendicular to that edge. That forces the normal into a vertical plane, where the angle between the face and the base can be read off directly. Then the angle between two faces comes from a dot product of normals, and the rest is careful trigonometric algebra plus one use of AM–GM.

## Solution

Let the centre of the base be the origin $O$, with $\mathbf{i}$ along $AB$, $\mathbf{j}$ along $BC$ and $\mathbf{k}$ up. Take

$$A=(-a,-b,0),\quad B=(a,-b,0),\quad C=(a,b,0),\quad D=(-a,b,0),\quad V=(0,0,h),$$

with $a,b,h>0$.




Let $M=(0,-b,0)$ be the midpoint of $AB$ and $N=(a,0,0)$ the midpoint of $BC$. Since $VA=VB$, the line $VM$ lies in face $AVB$ and is perpendicular to $AB$; so is $OM$ in the base. So the angle between the face and the base is angle $VMO$, and

$$\tan\alpha=\frac{h}{b},\qquad\text{and similarly}\qquad\tan\beta=\frac{h}{a}.$$

**Part (i)**

Face $AVB$ contains the directions $\overrightarrow{AB} \parallel \mathbf{i}$ and $\overrightarrow{MV} = b\,\mathbf{j} + h\,\mathbf{k}$. A normal is

$$\mathbf{i} \times (b\,\mathbf{j} + h\,\mathbf{k}) = -h\,\mathbf{j} + b\,\mathbf{k}.$$

Dividing by its length $\sqrt{h^2 + b^2}$ and using $\tan\alpha = h/b$ gives the unit normal

$$\mathbf{n}_1 = -\sin\alpha\,\mathbf{j} + \cos\alpha\,\mathbf{k}.$$

Its $\mathbf{j}$-component is negative, so it points out of the pyramid (face $AVB$ is on the side $y < 0$).

In exactly the same way, face $BVC$ contains $\mathbf{j}$ and $\overrightarrow{NV} = -a\,\mathbf{i} + h\,\mathbf{k}$, and its outward unit normal is

$$\mathbf{n}_2 = \sin\beta\,\mathbf{i} + \cos\beta\,\mathbf{k}.$$

Hence

$$\mathbf{n}_1 \cdot \mathbf{n}_2 = \cos\alpha\cos\beta.$$

**Ask yourself** Is the angle between the normals the angle between the faces, or its supplement?

Look at a cross-section perpendicular to the common edge $VB$. The two faces appear as two lines meeting at the interior angle $\pi - \theta$, and each outward normal is perpendicular to its line and points away from the solid. Turning one face onto the other through the interior angle turns the outward normal through the same angle, but in the cross-section the two normals both point *outwards*, so they make the angle $\pi - (\pi - \theta) = \theta$. (Picture a cube: faces at $\frac{\pi}{2}$, normals at $\frac{\pi}{2}$; flatten the pyramid and the faces approach angle $\pi$ while the normals approach angle 0.) Therefore

$$\cos\theta = \mathbf{n}_1 \cdot \mathbf{n}_2 = \cos\alpha\cos\beta,$$

as required. Since $0 < \cos\alpha\cos\beta < 1$, $\theta$ is acute, consistent with the faces meeting at an obtuse angle.

**Part (ii)**

The foot of the perpendicular from $V$ is $O$, so the angle between $BV$ and the base is angle $VBO$, with

$$\cot\phi = \frac{OB}{h} = \frac{\sqrt{a^2 + b^2}}{h}.$$

Therefore

$$\cot^2\phi = \frac{a^2}{h^2} + \frac{b^2}{h^2} = \cot^2\beta + \cot^2\alpha,$$

using $\cot\beta = a/h$ and $\cot\alpha = b/h$. This is the first result.

For the second, write $p = \cos^2\alpha$, $q = \cos^2\beta$, so that $\cos^2\theta = pq$ by part (i). Then

$$\cot^2\phi = \frac{p}{1-p} + \frac{q}{1-q} = \frac{p + q - 2pq}{(1-p)(1-q)}.$$

Since $\cos^2\phi = \dfrac{\cot^2\phi}{1+\cot^2\phi}$, we need

$$1+\cot^2\phi = \frac{(1-p)(1-q)+p+q-2pq}{(1-p)(1-q)} = \frac{1-pq}{(1-p)(1-q)},$$

and dividing,

$$\cos^2\phi = \frac{p+q-2pq}{1-pq} = \frac{\cos^2\alpha+\cos^2\beta-2\cos^2\theta}{1-\cos^2\theta}.$$

(The denominator is positive since $\cos\theta < 1$.)

Now $\cos\alpha$ and $\cos\beta$ are positive, so by the AM–GM inequality

$$\cos^2\alpha+\cos^2\beta \;\geq\; 2\cos\alpha\cos\beta = 2\cos\theta,$$

and since $1-\cos^2\theta > 0$,

$$\cos^2\phi \;\geq\; \frac{2\cos\theta-2\cos^2\theta}{1-\cos^2\theta}.$$

**Ask yourself** How does this lower bound tell us anything about $\phi$ and $\theta$?

Simplify it and compare it with $\cos^2\theta$. Cancelling the positive factor $1-\cos\theta$,

$$\frac{2\cos\theta(1-\cos\theta)}{(1-\cos\theta)(1+\cos\theta)} = \frac{2\cos\theta}{1+\cos\theta}.$$

Writing $c = \cos\theta \in (0,1)$,

$$\frac{2c}{1+c} - c^2 = \frac{c\,(2-c-c^2)}{1+c} = \frac{c\,(1-c)(2+c)}{1+c} > 0.$$

So $\cos^2\phi \geq \dfrac{2c}{1+c} > \cos^2\theta$. Both $\phi$ and $\theta$ lie in $(0, \frac{\pi}{2})$, where cosine is positive and decreasing, so $\cos\phi > \cos\theta$ gives $\phi < \theta$.

## Checks

- Square base with $h$ equal to half a side ($a = b = h$): $\alpha = \beta = \frac{\pi}{4}$, so $\cos\theta = \frac{1}{2}$ and $\theta = 60^\circ$. Directly, $\cot\phi = \sqrt{2}$, so $\cot^2\phi = 2 = 1+1$ and $\phi \approx 35.26^\circ < 60^\circ$. Also $\cos^2\phi = \frac{2}{3}$, matching $(\frac{1}{2}+\frac{1}{2}-\frac{1}{2})/(1-\frac{1}{4}) = \frac{2}{3}$.
- A computer check over $10^5$ random pyramids, measuring the angle between the faces directly from a cross-section through $VB$, agreed with $\cos\theta = \cos\alpha\cos\beta$ and with both formulas for $\phi$, and always gave $\phi < \theta$.

## Common slips

- Taking an inward normal for one face and an outward normal for the other: the dot product then has the wrong sign and gives $\cos(\pi-\theta)$.

- Measuring $\alpha$ along the edge $VA$ rather than along the slant height $VM$; the angle between planes must be measured perpendicular to their line of intersection.
- Stopping at $\cos^2\phi \geq \dfrac{2\cos\theta}{1+\cos\theta}$ without showing that this exceeds $\cos^2\theta$.

## Beyond STEP

The angle between two planes equals the angle between their normals (up to taking the supplement), which is the start of how normals are used to describe surfaces in multivariable calculus and differential geometry. The identity $\cos\theta = \cos\alpha\cos\beta$ is also a disguised form of the spherical Pythagoras theorem: the three normals $\mathbf{k}$, $\mathbf{n}_1$, $\mathbf{n}_2$ form a right-angled spherical triangle. Strang's *Introduction to Linear Algebra* is a good place to deepen your feel for dot and cross products.

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